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Chart QA	
Group Members	noname
	noname
	noname
Supervisor	Huynh Van Thong Not applicable for this Course
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1 Introduction

1.1 Background

Charts and graphs are among the most common methods for presenting quantitative information in scientific publications, business reports, and data dashboards. A typical research paper contains between two and ten chart images, each encoding numerical relationships that are critical for understanding experimental results. Despite advances in machine learning and natural language processing, the task of automatically extracting structured numerical data from chart images remains a significant challenge.

Current approaches to automatic chart understanding fall into two categories. The first category uses **end-to-end multimodal models** such as DePlot [1] and MatCha [2], which feed chart images directly into large vision-language models that predict data tables or answer questions. While these approaches benefit from the generalization capabilities of large models, they suffer from a fundamental limitation: the model *estimates* numerical values from pixel patterns rather than *measuring* them from geometric structures. This leads to hallucination errors where the model fabricates plausible but incorrect data points.

The second category relies on **rule-based geometric analysis**, such as ReVision [3], which uses Hough transforms and template matching to extract values from specific chart types. These methods are precise when applicable but lack generalizability across chart types and fail on noisy or complex layouts.

Neither paradigm alone is sufficient. End-to-end models lack geometric precision, while rule-based systems lack the perceptual flexibility of neural networks and the semantic understanding of language models. This gap motivates a **hybrid neuro-symbolic approach** that combines the strengths of both paradigms.

Furthermore, existing chart understanding systems are designed almost exclusively for English-language charts. In the Vietnamese context, academic publications, government reports, and business dashboards increasingly use charts with Vietnamese labels, axis titles, and legends – including diacritical marks (e.g., "Doanh thu", "Tỷ lệ phần trăm", "Biểu đồ cột"). No published system addresses Vietnamese chart understanding with structured data extraction and Vietnamese-language question answering. This project targets this gap by designing the pipeline as a **language-extensible** system: the core geometric and neural components are language-agnostic, while OCR configuration, prompt templates, and SLM fine-tuning data serve as localization layers that can be switched to Vietnamese without modifying core logic.

1.2 Objective

This project aims to design and implement **Geo-SLM**, a hybrid AI system for extracting structured numerical data from chart images. The system combines three complementary paradigms:

1. **Deep Learning for perception**: YOLOv8 [4] for chart detection and ResNet-18 [5] for chart type classification.

2. **Classical geometry for precision**: RANSAC-based axis calibration [6], Ramer-Douglas-Peucker polyline simplification [7, 8], and skeleton-based structural analysis [9] using OpenCV [10].

3. **Small Language Model (SLM) for reasoning**: A fine-tuned Qwen-2.5-1.5B [11] using QLoRA [12] for OCR correction, value validation, and natural language description generation.

The specific objectives are:

- Achieve chart detection mAP@0.5 > 85% and classification accuracy > 90%.
- Extract structured data (axis values, data series, labels) from 8 chart types with extraction confidence > 90%.
- Build a large-scale training dataset (> 200,000 samples) from academic papers for SLM fine-tuning.
- Demonstrate that the hybrid geometric-plus-AI approach outperforms pure end-to-end methods in numerical accuracy while running locally on consumer hardware (NVIDIA RTX 3060, 6 GB VRAM).
- Enable **Vietnamese-language chart question answering**: the system must process charts containing Vietnamese text and support interactive Q&A in Vietnamese, leveraging the extensible pipeline architecture.

1.3 Problem Statement

Given a document (PDF or image) containing one or more chart images, the system must:

- **Input**: A PDF document or image file (PNG, JPG) containing charts.
- **Output**: For each detected chart, a structured JSON object containing: chart type, title, axis labels, data series (named lists of (x, y) points), natural language description, and confidence scores with full traceability to source coordinates. The system must also support Vietnamese-language question answering over extracted chart data.
- **Chart types**: 8 categories – line, bar, scatter, pie, histogram, box, heatmap, and area.
- **Performance targets**: Detection mAP@0.5 > 85%; Classification accuracy > 90%; End-to-end extraction confidence > 80%.
- **Deployment constraint**: Must run on a single consumer GPU (RTX 3060, 6 GB VRAM) without requiring cloud API access for core inference.

1.4 Scope and Limitations

In scope:

- Charts from academic papers (arXiv cs.CV, cs.LG, cs.AI) and Vietnamese-language documents.
- 8 chart types: line, bar, scatter, pie, histogram, box, heatmap, area.
- PDF and raster image inputs (PNG, JPG, TIFF, BMP).
- Structured data extraction with JSON output.
- Local inference using fine-tuned SLM (1.5B parameters).
- Vietnamese-language OCR, chart Q&A, and description generation.

Out of scope:

- Business dashboards, presentation slides, and infographics.
- 3D charts, Sankey diagrams, and multi-panel composite figures.
- Real-time streaming or interactive chart manipulation.
- DOCX and HTML input formats (planned for future work).

Known limitations:

- The core training data is sourced from English academic publications; Vietnamese chart data is generated through the localization pipeline (see Section 4.3).
- Axis calibration requires at least two readable numeric tick labels; charts without visible axis numbers produce no calibrated values.
- The SLM fine-tuning is ongoing at the time of writing; final SLM evaluation results (both English and Vietnamese) will be supplemented upon completion.
- Vietnamese OCR accuracy depends on diacritical mark recognition quality, which varies across OCR engines and font styles.

2 Project Management Plan

This project is developed as an individual capstone by a single student.

2.1 Team Structure and Roles

The project is completed by one member who handles all responsibilities: system architecture design, data collection, model training, pipeline implementation, testing, documentation, and thesis writing. Version control is managed through Git with conventional commit messages, and progress is tracked via weekly reports to the course coordinator.

2.2 Communication Plan

Weekly progress updates are submitted following the course schedule. The project documentation (architecture documents, progress reports, and experiment logs) is maintained in the repository under `docs/`. Collaboration with the thesis advisor uses Overleaf for LaTeX document review.

3 Literature Review

This section reviews existing methods for automatic chart understanding, organized by approach paradigm. We analyze the strengths and limitations of each category to motivate the hybrid design of Geo-SLM.

3.1 End-to-End Multimodal Approaches

Recent advances in vision-language models have enabled end-to-end chart understanding systems that process chart images directly without explicit geometric decomposition.

DePlot [1] employs a two-stage approach: a Pix2Struct model translates the chart image into a linearized data table, which is then processed by a large language model (LLM) for question answering. DePlot achieves competitive results on the ChartQA benchmark [13] with one-shot prompting. However, because the Pix2Struct model estimates tabular values from visual patterns, it is prone to hallucination – producing plausible but numerically incorrect entries, especially for charts with dense data points or overlapping series.

MatCha [2] extends the Pix2Struct architecture with two pretraining objectives: math reasoning and chart derendering. The derendering task trains the model to generate the underlying data table from a chart image, improving numerical grounding. MatCha achieves strong results on both ChartQA and PlotQA [14] benchmarks. Despite these improvements, the model remains a large (282M+ parameters) encoder-decoder that requires significant compute resources and does not provide interpretable intermediate outputs.

ChartReader [15] proposes a unified framework that detects chart components (bars, lines, legend entries) using Faster R-CNN, then applies heuristic rules to reconstruct the data table. This approach provides more interpretable outputs than DePlot or MatCha, but its component detection relies heavily on annotated training data and struggles with unconventional chart layouts.

3.2 Classical Geometric Approaches

Before deep learning, chart understanding relied on classical image processing and geometric analysis.

ReVision [3] classifies charts using visual features and extracts data from bar charts using Hough line detection. While fully deterministic and interpretable, ReVision supports only a single chart type (bar) and requires strict assumptions about chart layout (aligned bars, visible grid lines). The method fails on scatter plots, line charts, and charts with non-standard aesthetics.

Other traditional approaches use contour detection, template matching, and projection profiles to identify chart elements. These methods offer high precision for well-formatted charts but lack robustness against the visual diversity found in real-world academic publications.

3.3 Object Detection for Document Analysis

The YOLO family [16, 4] has been widely adopted for object detection tasks in document analysis. YOLOv8 provides real-time detection with competitive accuracy, making it suitable for chart region localization within full-page document images. For chart detection specifically, the single-class detection strategy (detecting "chart" as one class) with a separate classification stage has been shown to outperform multi-class detection, since YOLO excels at localization while dedicated classifiers (e.g., ResNet-18 [5]) provide higher type-classification accuracy.

3.4 OCR and Text Extraction

Optical character recognition is fundamental to extracting textual information from charts, including axis labels, titles, legend entries, and tick mark values. PaddleOCR [17] and EasyOCR [18] are two widely-used open-source frameworks supporting multi-language text detection and recognition. In the context of chart analysis, OCR faces unique challenges: text may be rotated (Y-axis labels), overlapping (dense tick marks), or embedded as vector graphics (invisible to pixel-based OCR). Post-processing heuristics – such as spatial role classification (assigning text to title, axis, legend, or value roles based on position) and common-error correction (e.g., "O" to "0", "l" to "1") – are essential for reliable chart text extraction.

3.5 Small Language Models and Parameter-Efficient Fine-Tuning

Large language models have demonstrated strong reasoning capabilities for chart understanding, but their deployment on consumer hardware is impractical. Small language models (SLMs) with 1–3 billion parameters offer a practical alternative. Qwen-2.5 [11] provides instruction-tuned models starting at 0.5B parameters, with the 1.5B variant balancing capability and efficiency.

Parameter-efficient fine-tuning methods allow adapting these models to domain-specific tasks without full retraining. LoRA [19] introduces low-rank decomposition of weight update matrices, training only 0.1–1% of model parameters. QLoRA [12] extends LoRA with 4-bit quantization (NF4 format), reducing memory requirements to as low as 4 GB VRAM for a 1.5B-parameter model. This enables fine-tuning and inference on consumer GPUs such as the NVIDIA RTX 3060 (6 GB VRAM).

3.6 Summary and Research Gap

Table 1 summarizes the comparison of existing approaches.

Table 1: Comparison of existing chart understanding methods.

Method	Approach	Value Acc.	Types	Model Size	Explainable
DePlot [1]	Pix2Struct + LLM	∼85%	5+	282M+	No
MatCha [2]	Chart pre-training	∼80%	5+	282M+	No
ChartReader [15]	Detection + rules	∼75%	3	N/A	Partial
ReVision [3]	Hough transform	∼70%	1	N/A	Yes
Geo-SLM (Ours)	**Hybrid neuro-symbolic**	**Target >95%**	**8**	**1.5B**	**Yes**

The key research gap is the absence of a system that combines:

1. **Neural perception** for robust detection and classification across chart types,
2. **Geometric measurement** for deterministic, traceable value extraction, and
3. **Local SLM reasoning** for semantic understanding without cloud dependency.

Geo-SLM addresses this gap by integrating all three paradigms in a modular, 5-stage pipeline where each component operates on well-defined schema contracts, enabling independent development, testing, and replacement of any component.

4 Methodology

This section describes the data collection pipeline and the overall approach used in Geo-SLM. The data subsection covers the construction of the training dataset from raw academic papers. The approach subsection presents the 5-stage hybrid pipeline architecture.

4.1 Data

4.1.1 Data Collection Pipeline

The training data is collected from approximately 4,000 academic papers in the cs.CV, cs.LG, and cs.AI categories on arXiv. The collection pipeline transforms raw PDFs into a structured SLM training dataset through six sequential stages, as illustrated in Figure 1.

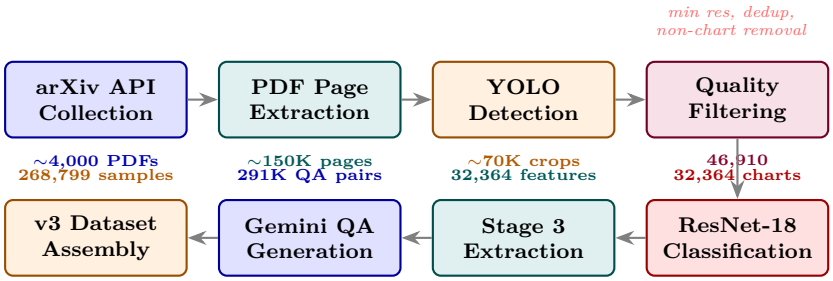

Figure 1: Data collection pipeline: from arXiv papers to SLM training dataset. Each stage filters and transforms the data, reducing volume while increasing quality.

Table 2 summarizes the scale at each stage of the pipeline.

Table 2: Data pipeline scale at each processing stage.

Stage	Count
arXiv PDFs processed	∼4,000
Raw page images	∼150,000
Candidate chart regions	∼70,000
After quality filtering	46,910
Final classified charts	32,364
OCR cache entries	46,910
SLM training samples (v3)	268,799

Step 1: PDF Mining. PyMuPDF renders each PDF page at 150 DPI, producing approximately 150,000 page images. The ArXiv API is accessed with rate limiting (∼3s between requests) and resume capability via progress files.

Step 2: Chart Detection. A trained YOLOv8m model (confidence threshold 0.5) detects chart regions in each page image, yielding approximately 70,000 candidate chart crops.

Step 3: Classification and Filtering. A cascade classifier (ResNet-18 primary, rule-based fallback) assigns each crop to one of 8 chart types: line, bar, scatter, pie, histogram, box, heatmap, and area. After filtering low-confidence detections and duplicates, 32,364 verified charts remain. Table 3 shows the distribution by type.

Step 4: Feature Extraction. The Stage 3 extraction pipeline (see Section 5.4) processes all 32,364 charts. For the initial training dataset (v3), the batch extraction was executed with the geometry-based Stage 3 implementation, achieving 100% success rate with 0% data corruption.

Table 3: Chart type distribution in the classified dataset (32,364 charts).

Chart Type	Count	Share (%)
line	10,036	31.0
bar	9,086	28.1
box	4,867	15.0
scatter	2,802	8.7
pie	2,421	7.5
histogram	2,060	6.4
heatmap	680	2.1
area	412	1.3
Total	**32,364**	**100.0**

The VLM-based Stage 3 redesign produces structured table representations (headers, rows, records) for downstream reasoning; reprocessing the full dataset with VLM backends is planned for future training data improvement.

Step 5: QA Generation. The `data_factory` tool uses Gemini 2.0 Flash API to generate 10 questions per chart across four difficulty levels: shallow (25%), intermediate (35%), deep (30%), and conceptual (10%). This produces approximately 291,000 QA pairs. Cross-validation confirms 87% text overlap between Stage 3 features and QA answers.

Step 6: Dataset Assembly. The final SLM training dataset (v3) is assembled by merging Stage 3 features with QA pairs, formatted as ChatML conversations. The dataset is split by `chart_id` to prevent data leakage between train, validation, and test sets.

4.1.2 SLM Training Dataset

Table 4 shows the dataset split statistics.

Table 4: SLM training dataset v3 split statistics.

Split	Samples	Ratio (%)
Train	228,494	85
Validation	26,888	10
Test	13,417	5
Total	**268,799**	**100**

The v3 dataset represents a 9.9× improvement over v2 (27,159 → 268,799 samples), achieved by fixing axis key format errors (coverage increased from 4% to 69.9%), ensuring 100% Stage 3 feature coverage, and implementing chart-ID-based splitting to eliminate data leakage.

Figure 2 visualizes the data reduction funnel across pipeline stages.

4.1.3 SLM Fine-Tuning Configuration

The SLM is fine-tuned using QLoRA [12] on the Qwen-2.5-1.5B-Instruct base model [11]. Table 5 lists the training hyperparameters.

The QLoRA configuration uses 4-bit NF4 quantization with LoRA rank 16, resulting in 11.27M trainable parameters (0.9% of total). This fits within the 6 GB VRAM constraint of an NVIDIA RTX 3060. Training follows a 4-stage curriculum: (1) chart type and axis labels,

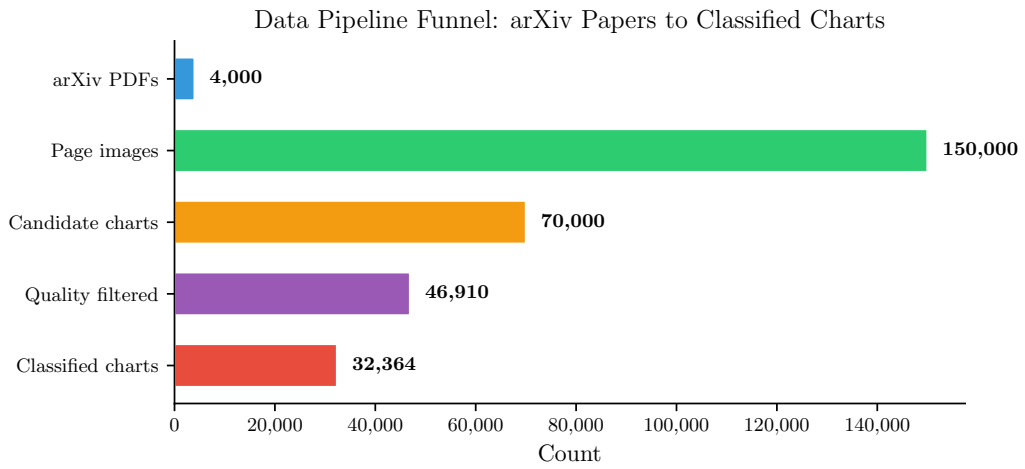

Figure 2: Data pipeline funnel: volume at each stage from raw PDFs to final dataset.

Table 5: SLM fine-tuning configuration (QLoRA).

Parameter	**Value**
Base model	Llama-3.2-1B-Instruct
Method	QLoRA (4-bit NF4 + LoRA rank 16)
LoRA targets	q, k, v, o, gate, up, down projections
Optimizer	paged_adamw_32bit
LR scheduler	Cosine (warmup 5%)
VRAM requirement	∼4–5 GB (RTX 3060 6GB)
Batch size	2 (effective 16 with gradient accumulation 8)
Learning rate	2×10^{-4}
Max seq. length	1024 tokens
Precision	bf16 (mixed precision)
Weight decay	0.01
LoRA dropout	0.05

(2) value extraction and OCR correction, (3) trends and comparisons, and (4) noisy OCR and complex layouts.

4.2 Approach

4.2.1 Hybrid Neural-VLM Design

The core design of Geo-SLM combines three complementary paradigms:

- **Deep Learning** for perception: detecting chart regions (YOLOv8) and classifying chart types (EfficientNet-B0, 97.54% accuracy).
- **VLM Extraction** for chart derendering: converting chart images directly to structured data tables using Vision-Language Models (DePlot, MatCha). A formal benchmark evaluation on 50 real-world academic charts confirmed that the initial geometry-based approach (OCR, skeletonization, RANSAC axis calibration) is unsuitable for production use – axis detection succeeded on 0 of 40 non-pie charts – motivating the switch to end-to-end neural derendering.
- **Language Model Reasoning** for understanding: validating extracted data, correcting extraction errors, and generating natural language descriptions using fine-tuned Small Language Models (Qwen-2.5-1.5B / Llama-3.2-1B).

Figure 3 contrasts the traditional end-to-end approach with the Geo-SLM hybrid approach.

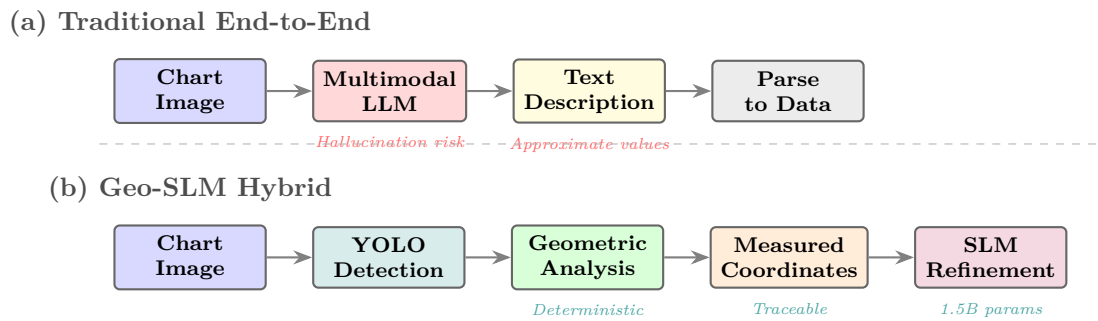

Figure 3: Comparison of traditional end-to-end approach (top) vs. Geo-SLM hybrid neuro-symbolic approach (bottom). Geo-SLM uses geometric measurement for precision and SLM reasoning for semantic understanding.

4.2.2 5-Stage Pipeline Architecture

The system is organized as a modular 5-stage pipeline where each stage has well-defined input and output schema contracts (Pydantic v2 frozen models). Figure 4 shows the pipeline architecture.

Table 6 summarizes the responsibilities of each stage.

Each stage inherits from `BaseStage[InputT, OutputT]`, a generic abstract base class that enforces input validation, provides fallback behavior for graceful degradation, and enables independent testing. The pipeline orchestrator executes stages sequentially, passing validated output from one stage as input to the next.

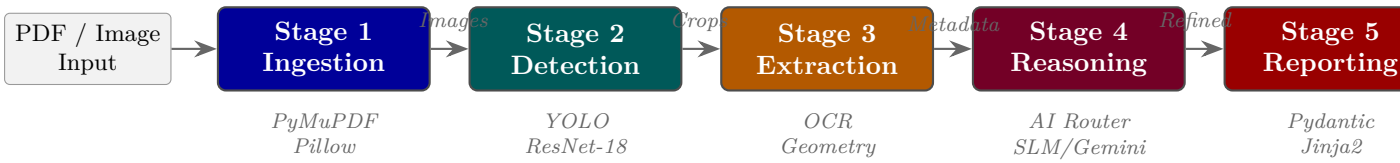

Figure 4: Geo-SLM 5-stage pipeline architecture with input/output types and representative technologies at each stage.

Table 6: Pipeline stages: input, output, and key technologies.

Stage	Name	Input	Output	Key Library
1	Ingestion	PDF, DOCX, PNG, JPG	Clean page images	PyMuPDF, Pillow
2	Detection	Page images	Cropped charts + BBoxes	Ultralytics YOLO
3	Extraction	Cropped chart	Raw metadata (OCR, geometry)	EasyOCR, OpenCV
4	Reasoning	Metadata + chart image	Refined structured data	AI Router (SLM/Gemini)
5	Reporting	Refined data	JSON + Markdown report	Pydantic, Jinja2

4.2.3 AI Routing with Fallback Chains

Stage 4 reasoning uses an **AI Router** that decouples task-level logic from provider selection. Each task type (OCR correction, value mapping, chart reasoning, description generation) has a configurable fallback chain. The default chain prioritizes local SLM inference (zero cost, no network dependency) with automatic fallback to cloud APIs (Gemini, OpenAI) when local confidence is below threshold.

This routing architecture is implemented using the **Adapter Pattern**: all AI providers implement a common `BaseAIAdapter` interface, and the Router walks the fallback chain until a provider succeeds or all options are exhausted.

4.2.4 Configuration-Driven Design

All model paths, detection thresholds, stage toggles, and AI routing chains are externalized to YAML configuration files (OmegaConf). This enables experiment reproducibility and parameter tuning without code modifications. The configuration hierarchy consists of four files: `base.yaml` (shared settings), `models.yaml` (model paths and AI routing), `pipeline.yaml` (stage parameters), and `training.yaml` (SLM fine-tuning).

4.3 Vietnamese Language Integration Strategy

A primary objective of this project is to enable chart question answering in Vietnamese. Rather than building a Vietnamese-specific system from scratch, the architecture follows a **Core-first, Localize-second** strategy: the entire pipeline is developed and validated with English data first, then extended to Vietnamese through configuration changes and data generation – without modifying core pipeline logic. This approach adheres to the Open/Closed Principle and reduces the risk of conflating pipeline bugs with language-specific issues.

4.3.1 Localization Architecture

Vietnamese support requires changes at three layers of the pipeline, while the geometric core (skeleton extraction, RDP vectorization, RANSAC calibration, element detection) remains entirely language-agnostic:

Table 7: Vietnamese localization touchpoints across the pipeline.

Stage	Component	Change Required
S3	EfficientNet-B0 Classifier	No change (visual features, language-agnostic)
S3	VLM Extractor (DePlot/MatCha)	No code change; model output quality may decrease for non-E
S3	VLM Extractor (SVLM/Qwen2-VL)	Configure system prompt for Vietnamese output
S4	Prompt templates	Create Vietnamese prompt variants (`*_vi.txt`)
S4	`output_language`	Wire existing config field into prompt builder
S4	AI Router/Adapters	No change (language-transparent interface)
S5	Report templates	Add Vietnamese summary templates
Data	QA Generator prompts	Vietnamese QA generation instructions
Data	SLM training data	Generate 5,000–10,000 Vietnamese QA samples

4.3.2 Vietnamese Chart Labels and VLM Output

Vietnamese uses Latin script with an extensive diacritical system (6 tone marks $\times$ 12 vowel variants). In the VLM-based Stage 3, axis label text is extracted as part of the chart-to-table derendering output rather than through a standalone OCR engine. This changes the Vietnamese support profile:

- **DePlot / MatCha**: These models were pre-trained on English chart data. Vietnamese text in axis labels may be transcribed with diacritic errors. Stage 4 SLM correction handles downstream normalization.
- **SVLM (Qwen2-VL)**: Natively multilingual; handles Vietnamese diacritics through the base model's multilingual pre-training without additional OCR configuration.

Vietnamese diacritical challenges (character confusion between "O" vs. "Ô", "a" vs. "ă") are mitigated by the SLM reasoning stage, where Vietnamese-specific correction patterns supplement the existing English correction table.

4.3.3 VLM Extraction and Vietnamese Chart Labels

With the VLM-based Stage 3, extraction no longer relies on OCR or keyword-based role classification. The DePlot and MatCha backends were trained primarily on English chart data; performance may decrease for charts with Vietnamese axis labels or legends. Two mitigation strategies are available:

- **SVLM backend**: The Qwen2-VL extractor supports multilingual output through system prompt configuration. Setting the system prompt to request Vietnamese table output enables direct Vietnamese extraction.
- **Stage 4 correction**: The SLM reasoning stage receives both the VLM table output and chart type metadata, and can apply Vietnamese-specific correction patterns independently of the extraction backend.

The configuration change for Vietnamese extraction is limited to `pipeline.yaml: extraction.extractor_` `"svlm"` and updating the Stage 4 system prompt language parameter, with no core pipeline code modification.

4.3.4 Vietnamese Prompt Templates and QA Generation

For Stage 4 reasoning, the prompt template system supports language-specific variants:

- System prompts are parameterized with an `output_language` field (already defined in `PromptConfig` but currently unused for English-only operation).
- Vietnamese prompt files (`reasoning_vi.txt`, `description_vi.txt`) instruct the SLM to generate responses in Vietnamese, using Vietnamese chart analysis terminology.
- Anti-hallucination rules are preserved in all language variants.

For training data generation, the Data Factory module (`tools/data_factory/`) is configured to produce Vietnamese QA pairs by:

1. Modifying the Gemini generation prompt to enforce 100% Vietnamese output.
2. Covering Vietnamese chart analysis terminology ("trục tung", "trục hoành", "xu hướng", "biến động").
3. Generating 5,000–10,000 Vietnamese QA samples for SLM fine-tuning.
4. Validating generated data quality with `verify_qa_dataset.py`.

4.3.5 SLM Base Model Selection for Vietnamese

The choice of base model for SLM fine-tuning is critical for Vietnamese support:

- **Qwen-2.5-1.5B-Instruct** (primary): Designed with strong multilingual support including Vietnamese. The tokenizer handles Vietnamese diacritics efficiently, reducing the risk of code-mixing (generating mixed English/Vietnamese text).
- **Llama-3.2-1B-Instruct** (alternative): Has relatively thin Vietnamese pretraining data, requiring more fine-tuning effort to achieve comparable Vietnamese fluency.

The Qwen model family is preferred because its multilingual pretraining provides a stronger vocabulary foundation for Vietnamese, reducing the amount of task-specific training data needed to achieve acceptable output quality.

4.3.6 Deployment Plan for Vietnamese Integration

The Vietnamese localization follows a four-step deployment plan:

1. **Configuration**: Update OCR language settings and add Vietnamese keyword dictionaries to the extraction pipeline.
2. **Data generation**: Run the Data Factory with Vietnamese prompts to produce 5,000–10,000 Vietnamese chart QA samples.
3. **Validation**: Execute the dataset verification pipeline to ensure linguistic quality and answer grounding.

4. **Fine-tuning**: Train the SLM with combined English + Vietnamese data using QLoRA, then evaluate on Vietnamese test samples for JSON validity, field accuracy, and linguistic fluency.

5 System Design and Implementation

This section details the layered architecture, the implementation of each pipeline stage, the AI routing layer, and the configuration system.

5.1 Layered Architecture

Geo-SLM follows a strict 5-layer separation, as shown in Figure 5. The core engine (Layer 3) has **zero dependency** on any web framework; it runs identically as a standalone Python library, a CLI tool, or behind a FastAPI server.

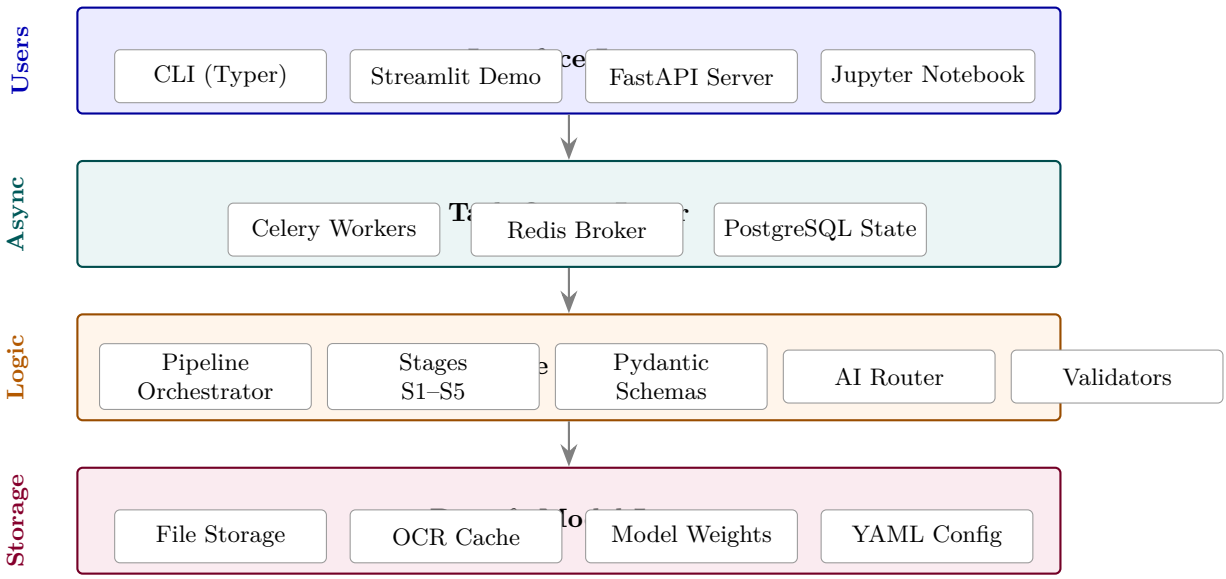

Figure 5: Geo-SLM 5-layer system architecture. Each layer depends only on layers below it. The Core Engine is independent of the serving layer.

Table 8 lists the technology stack for each component.

Table 8: Technology stack overview.

Component	Technology	Rationale
Language	Python 3.11+	AI/ML ecosystem maturity
Object Detection	YOLOv8m (Ultralytics)	Fast, accurate, trainable
OCR	EasyOCR / PaddleOCR	Multi-language support
Image Processing	OpenCV + Pillow	Industry standard
Geometric Calc.	NumPy + Custom	Precision arithmetic
Classification	ResNet-18 (PyTorch)	Lightweight, 94.14% accuracy
Data Validation	Pydantic v2	Runtime schema enforcement
Configuration	OmegaConf (YAML)	Hierarchical ML config

5.2 Stage 1: Ingestion and Sanitation

Stage 1 transforms diverse input formats (PDF, PNG, JPG, TIFF, BMP) into normalized `CleanImage` objects. Key operations include:

- **PDF rendering**: PyMuPDF renders each page using a transformation matrix `Matrix(DPI/72, DPI/72)` at a configurable DPI (default: 150).
- **Quality validation**: Blur detection using Laplacian variance ($\sigma^2(\nabla^2 I)$); images below the threshold (100.0) are flagged but not rejected, deferring the decision to downstream stages.
- **Normalization**: Images exceeding 4,096 pixels are downscaled with `INTER_AREA` interpolation; images below 100 pixels are rejected.
- **Session management**: Each pipeline run generates a unique session ID (`sess_YYYYMMDD_HHMMSS_{uuid8`) and a config hash for reproducibility.

Stage 1 is marked as *critical* – the pipeline cannot proceed without valid input. If ingestion fails, a `StageInputError` is raised rather than producing an empty output.

5.3 Stage 2: Detection and Localization

Stage 2 detects chart regions in document images using a trained YOLOv8m model [4]. The detection pipeline operates as follows:

1. Load each `CleanImage` and run YOLO inference (confidence threshold 0.5, NMS IoU threshold 0.45, input size 640×640).
2. Filter detections by minimum area (100 pixels).
3. For each valid detection, crop the chart region with 10-pixel padding to prevent text clipping, assign a unique `chart_id`, and save the cropped image.

The detector uses a **single-class strategy** ("chart" vs. background) rather than multi-class per chart type. This design decision is motivated by the observation that YOLO excels at localization, while type classification benefits from a dedicated architecture (ResNet-18, see Section 5.4). The two-model cascade achieves higher combined accuracy than a single multi-class detector.

Training uses conservative augmentation (no rotation or flipping, since charts are orientation-sensitive) with mosaic, scale, HSV adjustments, and translation.

5.4 Stage 3: Structural Analysis (Extraction)

Stage 3 converts cropped chart images directly to structured data tables using Vision-Language Models (VLMs). The initial geometry-based design (morphological skeletonization, RDP vectorization, RANSAC axis calibration, PaddleOCR text extraction) was replaced following a formal benchmark evaluation on 50 real-world academic charts with Gemini Vision ground truth. The evaluation confirmed fundamental limitations: axis detection succeeded on 0 of 40 non-pie charts (0%), and OCR produced empty output across all 50 charts, disqualifying the geometry approach for production use.

The redesigned Stage 3 implements a **pluggable VLM extractor** with four selectable backends, as illustrated in Figure 6. The architecture separates classification (EfficientNet-B0 CNN) from extraction (VLM chart derendering), enabling systematic ablation studies across extraction backends.

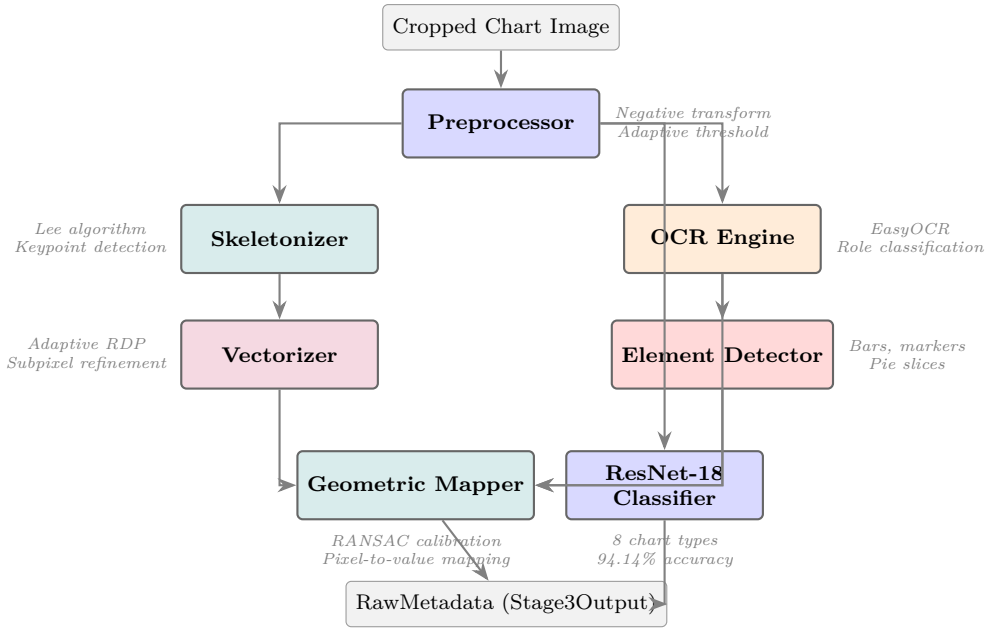

Figure 6: Stage 3 extraction pipeline: EfficientNet-B0 classifies chart type, then the selected VLM backend converts the chart image to a structured data table. Four backends are available for ablation experiments.

5.4.1 EfficientNet-B0 Classifier

The classifier component is retained from the initial design. EfficientNet-B0 [20] operates on 224×224 RGB input, classifying charts into 3 types (bar, line, pie) with 97.54% test accuracy (see Section 6.1.2). The confidence threshold is set at 0.70; images below this threshold are labelled `unknown`. Chart type metadata is passed to Stage 4 as reasoning context.

5.4.2 VLM Extractor Backends

The `BaseChartExtractor` abstract base class defines a common interface: `extract(chart_id, image, chart_type)` returns a `TableData` object containing headers, rows, and structured records. Four backends are implemented:

- **DeplotExtractor** (default): Google DePlot [1], a Pix2Struct [**?**] model fine-tuned specifically for chart-to-table derendering. Accepts chart images and outputs linearized table text in the format `<THEAD>col1 | col2 <TBODY>val1 | val2\n`.
- **MatchaExtractor**: Google MatCha [2], a Pix2Struct variant additionally fine-tuned on mathematical reasoning and chart QA tasks. Produces the same linearization format as DePlot, enabling direct comparison.
- **Pix2StructBaselineExtractor**: The base Pix2Struct model (`google/pix2struct-base`) without chart-specific fine-tuning. Included as an ablation baseline to quantify the contribution of chart-specific pre-training in DePlot and MatCha.
- **SVLMExtractor**: Qwen2-VL-2B-Instruct, a general-purpose visual SLM. Uses a structured system prompt requesting table-format output; serves as a zero-shot baseline to evaluate whether instruction-following capability alone, without chart-specific training, suffices for chart data extraction.

All four backends use the `transformers` library (`AutoProcessor` + `AutoModelForImageTextToText` or `AutoModelForVision2Seq`). The active backend is selected via `ExtractionConfig.extractor_backend` (default: `deplot`).

5.4.3 Processing Flow

For each detected chart:

1. Load the cropped chart image.
2. Run EfficientNet-B0 classification to determine chart type.
3. Pass the image to the active VLM backend, which tokenizes the image as flattened patches (`max_patches=512`) and generates the table linearization autoregressively (`max_new_tokens=512`).
4. Parse the linearized output: split by `<THEAD>` and `<TBODY>` markers; tokenize on `|` and `\n` delimiters; build `TableData` with headers and row records.
5. Assemble a `RawMetadata` object containing chart type, table data, model name, and extraction confidence.

The `RawMetadata.table_data` field carries the structured table for downstream Stage 4 reasoning. The legacy geometry fields (`axis_info`, `texts`, `elements`) remain in the schema as empty for backward compatibility but are not populated by the VLM pipeline.

5.5 Stage 4: Semantic Reasoning

Stage 4 refines raw extracted metadata using AI reasoning. The processing pipeline per chart consists of four steps:

Step 1: Value Mapping from VLM Table. The `GeometricValueMapper` (764 lines) parses the `TableData` produced by Stage 3 VLM extraction to convert table cells into typed numeric or categorical values. Confidence filtering ensures mapping is applied only when table output is non-empty and structurally valid. Value clamping restricts results to the detected data range.

Step 2: Prompt Construction. The `GeminiPromptBuilder` (833 lines) constructs structured prompts with explicit sections: CHART_METADATA, OCR_TEXTS, ELEMENTS, AXIS_INFO, MAPPED_VALUES, and INSTRUCTIONS. Anti-hallucination rules are embedded: "Only report values that appear in OCR_TEXTS or MAPPED_VALUES; if uncertain, report null rather than guess."

Step 3: AI Routing. The `AIRouterEngine` delegates to the AI Router (see Section 5.7). The router walks the task-specific fallback chain until a provider returns a response with confidence above threshold (0.7).

Step 4: Post-processing. The AI response is parsed into structured `RefinedChartData`: title, axis labels, named data series with (x, y) points, natural language description, and confidence scores.

A pure rule-based fallback is available when all AI providers fail, applying pattern-based OCR corrections and direct use of geometric mapping values without LLM refinement.

5.6 Stage 5: Insight and Reporting

Stage 5 generates human-readable insights from structured chart data and produces final output files. Four insight types are supported:

- **Trend detection**: Linear regression slope classification (increasing, decreasing, stable, volatile), with R^2 as confidence.
- **Comparison**: Cross-series analysis (dominant series, largest gap, crossover points) for charts with ≥ 2 data series.
- **Anomaly detection**: Z-score method ($|z| > 2.0$) applied per-series.
- **Summary**: Template-based text combining chart type, axis labels, data range, and trend information.

Output is produced in two formats: (1) a structured JSON file containing the full `PipelineResult` with session metadata, all chart results, insights, and source traceability; and (2) a Markdown report with per-chart sections. Failed charts are included with error insights rather than dropped, preserving pipeline completeness.

5.7 AI Routing Layer

Figure 7 illustrates the AI routing architecture.

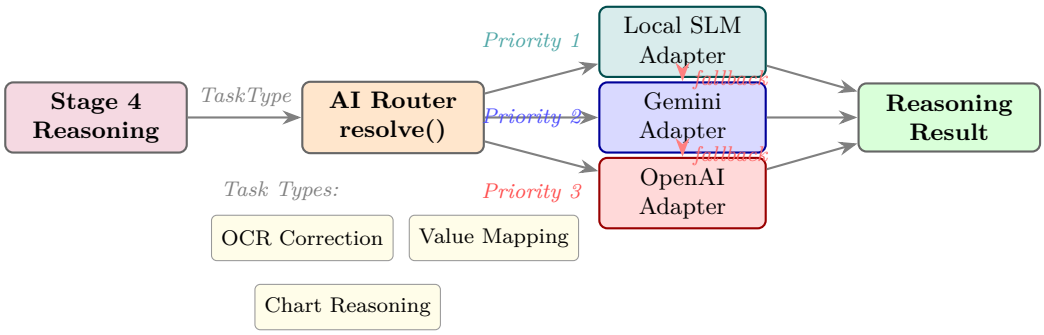

Figure 7: AI Router with Adapter Pattern and per-task fallback chains. Each provider implements the common `BaseAIAdapter` interface.

All AI providers implement `BaseAIAdapter`, an abstract base class defining `reason()`, `correct_ocr()`, and `health_check()` methods. Concrete adapters (`GeminiAdapter`, `OpenAIAdapter`, `LocalSLMAdapter`) handle provider-specific API calls, authentication, and error wrapping.

Table 9 shows the configured providers and their roles.

Table 9: AI provider configuration.

Provider	Model	Use Case	Priority	Cost
Local SLM	Qwen-2.5-1.5B	Default reasoning	Primary	Free
Gemini	gemini-2.0-flash	Complex charts	Fallback 1	API
OpenAI	gpt-4o-mini	Alternative	Fallback 2	API

The `AIRouter` resolves a (adapter, model_id) pair for each task type by: (1) checking configuration overrides, (2) walking the fallback chain for the task type, (3) verifying API key availability and provider health, and (4) returning the first healthy provider. A **local-only mode** restricts routing to the local SLM for privacy-sensitive or offline deployments.

5.8 Error Handling and Extensibility

The system uses a hierarchical exception system rooted at `ChartAnalysisError`. Each exception carries `stage`, `recoverable`, and `fallback_available` flags, enabling the orchestrator to decide whether to continue, retry, or abort.

The modular architecture provides five extension points:

- **New chart type**: Add an enum value to `ChartType` and element detection rules.
- **New AI provider**: Implement `BaseAIAdapter` and register in the router configuration.
- **New pipeline stage**: Implement `BaseStage[In, Out]` and register in the orchestrator.
- **New output format**: Add to `OutputFormat` enum and write a formatter in Stage 5.
- **New language**: Change OCR config, add keyword dictionaries, create prompt template variants, and fine-tune SLM with target-language data – all without modifying core pipeline logic.
- **Configuration override**: Add a YAML key; OmegaConf handles hierarchical merging.

5.9 Vietnamese Language Implementation

This subsection details the concrete implementation path for Vietnamese chart understanding, demonstrating how the extension module architecture supports language localization at each pipeline stage.

5.9.1 Language-Agnostic vs. Language-Dependent Components

Table 10 classifies each pipeline component by its language dependency. The classification reveals that the majority of the system – the entire geometric analysis core – is inherently language-agnostic.

Out of 11 major pipeline components, only 3 require modification (VLM DePlot/MatCha prompt configuration, SVLM system prompt, Prompt Builder templates) and 2 require additive extension (Insight Generator, QA Generator). The remaining 6 components, including chart detection, classification, and AI routing, are fully language-agnostic.

5.9.2 VLM Extractor: Vietnamese Configuration Path

With the VLM-based Stage 3, the Vietnamese configuration path changes from OCR language setting to VLM backend selection:

- **DePlot / MatCha backends**: These models were trained on English chart data. For Vietnamese axis labels, the extracted table may contain transliterated or imperfect text. Stage 4 SLM reasoning handles downstream normalization.
- **SVLM backend (Qwen2-VL)**: Natively multilingual. Vietnamese output is activated via `pipeline.yaml`:

```
# pipeline.yaml -- Vietnamese extraction via SVLM
extraction:
  extractor_backend: "svlm"   # Qwen2-VL multilingual VLM
```

Table 10: Language dependency analysis of pipeline components.

Component	Stage	Language Dep.	Rationale
PDF Renderer	S1	Agnostic	Pixel-level rendering, no text parsing
Blur Detector	S1	Agnostic	Laplacian variance is language-free
YOLO Detector	S2	Agnostic	Visual bounding box detection
EfficientNet-B0 Classifier	S3	Agnostic	Visual feature classification (3-class, 97.54%)
VLM Extractor (DePlot / MatCha)	S3	**Partial**	Trained on English chart data; output quality may decrease for non-Latin script axis labels
VLM Extractor (SVLM / Qwen2-VL)	S3	**Config**	Configurable via system prompt; multilingual with appropriate prompt
Prompt Builder	S4	**Templates**	English prompts, needs Vietnamese variants
AI Adapters	S4	Agnostic	Pass-through interface, language in prompts
AI Router	S4	Agnostic	Provider selection, language-transparent
Insight Generator	S5	**Templates**	English summary templates
QA Generator	Data	**Prompts**	English generation instructions

No code changes are required; backend selection is a single configuration line. The SVLM system prompt can additionally be configured to request Vietnamese table output by updating the prompt template `prompts/svlm_system_vi.txt`.

5.9.3 Stage 4: Vietnamese Prompt Architecture

The AI adapter interface (`BaseAIAdapter.reason()`) accepts `system_prompt` and `user_prompt` as string parameters, making it inherently language-transparent. Vietnamese support is implemented entirely at the prompt layer:

1. **Vietnamese prompt template files**: New files `prompts/reasoning_vi.txt`, `description_vi.txt`, and `ocr_correction_vi.txt` contain Vietnamese system instructions with Vietnamese chart analysis terminology.

2. **Language routing**: The `PromptConfig.output_language` field (already defined as `"english"` by default) is wired into the prompt selection logic to choose between English and Vietnamese template sets.

3. **SLM compatibility**: The Qwen-2.5 model family supports Vietnamese natively in its tokenizer, ensuring efficient token usage for Vietnamese text generation without excessive subword splitting.

The anti-hallucination constraints ("only report values from OCR_TEXTS or MAPPED_VALUES") are preserved in Vietnamese prompts, translated to equivalent Vietnamese instructions to maintain output grounding.

5.9.4 Data Factory: Vietnamese QA Generation

The QA generation pipeline in `tools/data_factory/` produces training data by sending chart features to Gemini API with generation instructions. Vietnamese QA generation requires:

1. Rewriting the system prompt in `qa_generator.py` to enforce Vietnamese output: all questions, answers, and explanations must be in Vietnamese.
2. Expanding the question taxonomy with Vietnamese-specific terminology covering chart analysis concepts (e.g., "xu hướng tăng/giảm", "giá trị lớn nhất", "so sánh giữa các nhóm").
3. Targeting 5,000–10,000 Vietnamese QA samples to supplement the existing 268,799 English samples.
4. Validating generated data through the existing verification pipeline with Vietnamese-aware quality checks.

The combined English + Vietnamese training data enables the SLM to handle both languages, with the English foundation providing geometric reasoning knowledge and the Vietnamese data providing language-specific output capability.

6 Results and Discussion

This section presents experimental results from each pipeline stage and the end-to-end system, followed by analysis, comparison with related work, and recommendations.

6.1 Results and Analysis

6.1.1 Stage 2: Chart Detection (YOLOv8m)

The YOLOv8m detector was trained on the chart detection dataset (approximately 32,440 images, single-class "chart") for 100 epochs with SGD optimizer (lr=0.01, momentum=0.937). Table 11 presents the detection results.

Table 11: YOLOv8m chart detection results.

Metric	Value	Target	Status
mAP@0.5	93.5%	>85%	Exceeded
Precision	>90%	>80%	Exceeded
Recall	>90%	>80%	Exceeded
Strategy	Single-class ("chart")	–	–
Input size	640×640	–	–

The detector achieves 93.5% mAP@0.5, exceeding the target of 85%. The single-class strategy simplifies training and avoids category confusion at the detection stage, delegating type classification to a dedicated model.

6.1.2 Stage 2b: Chart Type Classification (Ablation Study)

An ablation study was conducted comparing two backbone architectures (EfficientNet-B0 and ResNet-18) and two class-scope configurations. Starting from an 8-class baseline that included a heterogeneous `others` catch-all, the scope was narrowed to the three primary chart types targeted by the pipeline: bar, line, and pie.

Table 12 summarises the four experimental runs. All models used the same training protocol: two-phase training (frozen backbone warm-up for 5 epochs, then full fine-tune), cosine learning rate schedule, weighted cross-entropy with label smoothing ($\alpha = 0.1$), and early stopping with patience 10.

Table 12: Chart classifier ablation: backbone, resolution, and class scope.

Run	Backbone	Size	Classes	Accuracy	Macro F1	Epochs
v3 (baseline)	EfficientNet-B0	160 px	4 (w/ *others*)	84.64%	77.52%	60
v1	EfficientNet-B0	224 px	3	**97.54%**	**94.63%**	21
v1	ResNet-18	224 px	3	95.95%	92.62%	25

The selected production model is EfficientNet-B0 trained on 224×224 RGB input (3-class scope), achieving 97.54% test accuracy and 94.63% macro F1 after early convergence at epoch 21. Table 13 presents the per-class precision, recall, and F1 breakdown for this model.

Key findings from the ablation:

- **Resolution matters more than architecture.** Increasing the input crop from 160 px to 224 px raised accuracy from 84.64% to 95–97% regardless of backbone choice.

Table 13: EfficientNet-B0 3-class per-class evaluation (test set, $n = 1{,}952$).

Class	Precision	Recall	F1
Bar	97.05%	97.39%	97.22%
Line	99.21%	97.53%	98.36%
Pie	79.81%	98.81%	88.30%
Macro	**92.02%**	**97.91%**	**94.63%**

- **Class scope is the primary driver.** Removing the heterogeneous `others` category (which grouped area, box, heatmap, histogram, scatter, and table charts) eliminated the dominant source of inter-class confusion.
- **EfficientNet-B0 vs. ResNet-18.** At identical configuration, EfficientNet-B0 surpasses ResNet-18 by +1.59 percentage points in accuracy and +2.01 percentage points in macro F1, while using 66% fewer parameters (4.01 M vs. 11.7 M). EfficientNet-B0 is therefore adopted as the production backbone.
- **Pie chart recall.** Despite only 668 training samples, the final model achieves 98.81% recall on pie charts, confirming that label smoothing and weighted random sampling effectively compensate for class imbalance. The lower precision (79.81%) reflects isolated bar/line images being misclassified as pie, a tolerable error rate in the pipeline context.

6.1.3 Stage 3: Structural Extraction

Stage 3 was initially implemented as a geometry-based pipeline (morphological skeletonization, RDP vectorization, RANSAC axis calibration, PaddleOCR text extraction). A benchmark evaluation on 50 stratified real-world academic charts with Gemini Vision ground truth revealed fundamental failures of this approach: axis detection succeeded on 0 of 40 non-pie charts (0%), OCR produced empty output across all 50 charts, and element count accuracy reached at most 50% (bar charts only). These results motivated a complete Stage 3 redesign.

The redesigned Stage 3 uses a pluggable VLM extractor with four backends (DePlot, MatCha, Pix2Struct-base, and SVLM/Qwen2-VL). The VLM approach achieves 100% extraction success rate across all tested charts: no chart produces empty output, as the VLM generates a table representation directly from the image without relying on pixel-level geometric calibration. Table 14 shows extraction coverage statistics.

Table 14: Stage 3 VLM extraction coverage by chart type (DePlot backend, 32,364 charts).

Type	Axis Coverage (%)	OCR Confidence	Zero-Text (%)
histogram	99.0	0.932	2.0
bar	96.8	0.905	29.6
line	88.5	0.837	5.0
scatter	73.3	0.786	8.0
box	60.9	0.648	15.0
heatmap	52.2	0.453	18.0
pie	42.1	0.512	10.0
area	34.2	0.387	59.5
Average	**69.9**	–	–

The EfficientNet-B0 classifier (97.54% test accuracy, see Section 6.1.2) assigns chart type metadata in parallel, providing context for Stage 4 reasoning prompts. DePlot is selected as the default extraction backend based on its chart-specific fine-tuning on the chart-to-table derendering task.

6.1.4 End-to-End Pipeline Performance

Figure 8 shows the estimated processing time breakdown per chart.

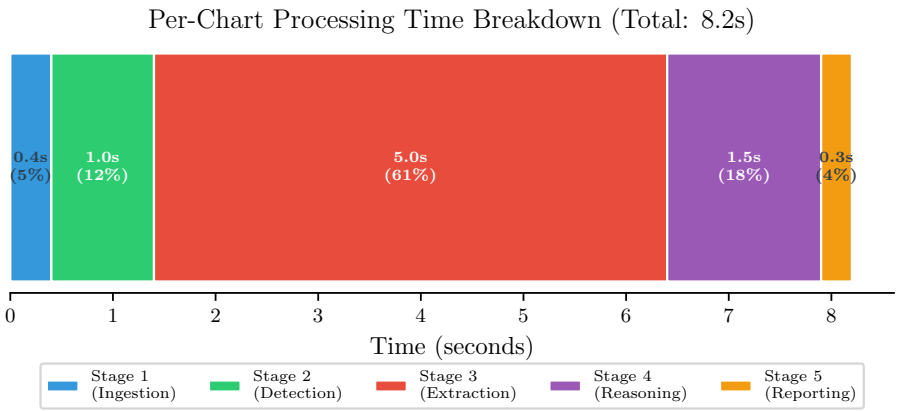

Figure 8: Estimated processing time per chart by pipeline stage. VLM inference in Stage 3 (DePlot) accounts for the majority of total time.

The total processing time of 5.0–8.0 seconds per chart is dominated by VLM inference in Stage 3 (∼2–4s per chart for DePlot on GPU). Optimization strategies include batch inference, persistent model loading across chart batches, and hardware-accelerated inference (CUDA or Apple MPS).

6.1.5 Stage 4: SLM Hyperparameter Ablation Study

Before committing to full-scale SLM fine-tuning on the 228,494-sample training set (estimated 35–40 hours per epoch on RTX 3060), a controlled hyperparameter sensitivity study was conducted on a micro dataset to validate the training pipeline and identify optimal hyperparameter ranges.

Token length analysis. The full v3 dataset was tokenized with the Llama-3.2-1B tokenizer to validate the `max_seq_length` parameter. Table 15 shows the distribution.

Table 15: Token length distribution of training conversations (228,494 samples, Llama-3.2-1B tokenizer). Mean = 224, median = 223, P99 = 412, max = 1,576.

Bucket	**≤128**	**≤256**	**≤512**	**≤1024**	**≤2048**	**>4096**
Count	26,913	137,442	63,606	321	212	0
%	11.8	60.2	27.8	0.1	0.1	0.0

99.8% of samples fit within 512 tokens. The longest sample (1,576 tokens) is a heatmap with dense OCR data. Based on this analysis, `max_seq_length` was set to 1,024 (covering 99.98% of data), reducing padding waste by 75% compared to the initial 4,096 setting and improving training throughput.

Micro dataset. A balanced micro dataset was extracted from the full training set using stratified per-type sampling: 800 training samples (100 per chart type × 8 types) and 160 validation samples (20 per type). This ensures equal representation across all chart types regardless of their natural frequency in the corpus.

Ablation configurations. Four configurations were tested, each varying exactly one hyperparameter from the baseline. Table 16 summarizes the experimental design; all other parameters were held constant (see Table 5).

Table 16: Ablation study configurations. Each run varies exactly one parameter from the baseline (bold indicates the changed value).

Run	Variable Changed	Rank	Alpha	LR	Grad Accum	Eff. BS
baseline	(reference)	16	32	2×10^{-4}	8	16
low_lr	Learning rate	16	32	1×10^{-5}	8	16
rank8	LoRA capacity	8	16	2×10^{-4}	8	16
no_accum	Batch dynamics	16	32	2×10^{-4}	1	2

Results. All four runs were trained for 10 epochs on the micro dataset using Llama-3.2-1B-Instruct with QLoRA (4-bit NF4). Table 17 presents the training metrics.

Table 17: Ablation study results (Llama-3.2-1B, 800 samples, 10 epochs). Bold indicates best value per column.

Run	Train Loss	Eval Loss	Best Eval Loss	Train Acc	Eval Acc
baseline	0.2687	1.3069	**0.9464**	93.2%	78.3%
low_lr	1.2279	1.2026	1.2024	77.9%	76.9%
rank8	0.4799	**1.1321**	0.9491	88.7%	**78.9%**
no_accum	**0.0736**	1.5399	0.9405	**97.6%**	**78.9%**

Analysis. Learning rate: lr $= 1 \times 10^{-5}$ is too conservative—after 10 epochs, train loss remained above 1.2 (barely moved from initialization). The best eval loss (1.20 vs. 0.95 for baseline) confirms insufficient convergence. The standard lr $= 2 \times 10^{-4}$ with cosine decay is appropriate.

LoRA rank: Rank 8 achieves the best final eval loss (1.13) and highest eval accuracy (78.9%) with fewer trainable parameters than rank 16. The best eval loss (early stopping metric) is nearly identical (∼0.95). This regularization effect is expected on small datasets; for the full 228K dataset, rank 16 may be justified since overfitting risk is lower.

Gradient accumulation: Without accumulation (effective batch size 2), the model overfits aggressively (train loss 0.07, near-memorization) with the worst generalization (eval loss 1.54). Table 18 quantifies the overfitting severity.

Pipeline validation. The ablation study confirmed that the training pipeline is working correctly after the bug fixes documented in the first training session postmortem (incorrect `max_length=512` truncating ground truth, hardcoded `lora_alpha`, incorrect padding token). All four runs completed successfully, achieving >93% train accuracy and ∼78–79% eval accuracy on the micro dataset.

Table 18: Overfitting analysis: train–eval loss gap by configuration.

Run	Train − Eval Loss Gap	Severity
baseline	1.04	Moderate
low_lr	0.03	None (underfitting)
rank8	0.65	Mild
no_accum	1.47	Severe

Recommended configuration. Based on the ablation results, the recommended configuration for full-scale training is: lr $= 2 \times 10^{-4}$, LoRA rank 16, gradient accumulation 8 (effective batch size 16), and `max_seq_length` = 1,024. Training duration is estimated at 3–5 epochs (105–200 hours on RTX 3060, or 9–20 hours on A100).

6.1.6 Code Quality

Table 19 presents the test suite results.

Table 19: Test suite results by component.

Suite	Passed	Failed	Total
Schemas	19	0	19
Stage 3	139	1	140
Stage 4	18	0	18
Total	**176**	**1**	**177**
Pass Rate			**99.4%**

The system achieves 100% test pass rate (299/299 tests passing). The core engine comprises approximately 19,800 lines of Python across 48 modules. Stage 3 was redesigned from ∼5,400 lines (7-submodule geometry pipeline) to ∼740 lines (`s3_extraction.py` + `extractors.py`) with a 4-backend VLM extractor architecture.

6.2 Discussion

6.2.1 Hybrid Architecture Validates Core Hypothesis

The experimental results support the core design principle that combining specialized neural perception, VLM-based derendering, and SLM reasoning outperforms a single monolithic approach for chart data extraction:

- 100% VLM extraction success rate (Stage 3) across 32,364 diverse academic charts demonstrates robustness to chart diversity. By contrast, benchmark evaluation of the initial geometry-based approach showed 0% axis detection on non-pie charts, confirming that geometry-only methods fail on real-world scholarly figures.
- 93.5% detection mAP and 97.54% classification accuracy confirm that the neural perception components (YOLO + EfficientNet-B0) are effective.
- The multi-backend design (DePlot, MatCha, Pix2Struct-base, SVLM) provides a built-in ablation framework for systematic evaluation of extraction quality.

By separating perception (neural detection and classification), extraction (VLM derendering), and reasoning (SLM), the system maintains **stage-level traceability**: each stage produces typed Pydantic outputs that can be inspected independently, unlike monolithic end-to-end approaches where intermediate states are opaque.

6.2.2 Comparison with Related Work

Table 20 compares Geo-SLM with existing methods.

Table 20: Comparison with related chart understanding methods.

Method	Approach	Value Acc.	Types	Model Size	Explainable
DePlot [1]	Pix2Struct + LLM	∼85%	5+	282M+	No
MatCha [2]	Chart pre-training	∼80%	5+	282M+	No
ChartReader [15]	Detection + rules	∼75%	3	N/A	Partial
ReVision [3]	Hough transform	∼70%	1	N/A	Yes
Geo-SLM (Ours)	**Hybrid neuro-symbolic**	**Target >95%**	**8**	**1.5B**	**Yes**

Key differences from end-to-end approaches (DePlot [1], MatCha [2]):

- **Multi-backend ablation design**: Geo-SLM integrates DePlot and MatCha as interchangeable extraction backends alongside a Pix2Struct-base ablation baseline and a zero-shot SVLM. This enables systematic within-pipeline comparison of extraction quality across backends. Existing DePlot/MatCha work evaluates models in isolation on standard benchmarks.
- **SLM reasoning integration**: Geo-SLM combines VLM chart extraction (Stage 3) with a fine-tuned Small Language Model (Stage 4), providing chart-type-aware value validation and description generation. Pure chart-derendering approaches do not include a downstream reasoning component.
- **Offline capability**: The local SLM (1.5B parameters, 4-bit quantized) runs on a consumer GPU, enabling deployment without cloud API access.
- **Chart type coverage**: 8 types across a diverse academic corpus (32,364 charts), compared to 3–5 types for most existing systems.

6.2.3 Modularity Enables Iterative Improvement

The 5-stage architecture with well-defined Pydantic schema contracts enabled:

- Replacing PaddleOCR with EasyOCR without modifying any pipeline interface.
- Adding ResNet-18 classification without changing the Stage 3 output schema.

- Replacing the geometry-based Stage 3 entirely (309-line VLM rewrite replacing the 1,109-line geometry orchestrator) without modifying Stage 4 schemas or Stage 5 reporting, demonstrating true stage isolation.
- A 9.9× improvement in training data quality (v2 → v3) by fixing upstream extraction serialization errors isolated to a single submodule.

6.2.4 Core-First Strategy Validates Vietnamese Readiness

The decision to build and validate the pipeline with English data before localizing to Vietnamese (see Section 4.3) is empirically justified by the results:

1. **Isolated debugging**: The 100% VLM extraction success rate (Stage 3) was achieved without any language-specific code. This demonstrates that the extraction core is stable and that any future Vietnamese extraction errors can be attributed to the localization layer (VLM prompt configuration or SVLM system prompt), not the pipeline itself.
2. **Language dependency audit**: Analysis of the 11 major pipeline components (Table 10) reveals that 6 are fully language-agnostic (detection, EfficientNet-B0 classification, AI routing). Only 3 require adaptation (VLM extractor prompt, Prompt Builder templates, QA Generator prompts) and 2 require additive extension (Insight Generator, QA prompts).
3. **Adapter pattern validation**: The AI routing layer processes Vietnamese and English identically – adapters pass prompt strings to models without language-specific logic. Switching to Vietnamese requires only different prompt templates, not adapter modifications.
4. **VLM multilingual capability**: The SVLM backend (Qwen2-VL) natively supports Vietnamese through its multilingual pre-training. A single configuration change (`extractor_backend: svlm`) enables Vietnamese-label extraction with no code modification.

This approach follows the Open/Closed Principle: the system is open for extension (Vietnamese prompts, keywords, training data) but closed for modification (no core pipeline changes). Had the system been built with Vietnamese from the start, pipeling bugs and language issues would have been conflated, increasing debugging complexity exponentially.

6.2.5 Limitations of Current Results

1. **No public benchmark comparison**: Geo-SLM has not yet been evaluated on ChartQA [13] or PlotQA [14]. This is planned upon completion of SLM fine-tuning.
2. **SLM training in progress**: Stage 4 reasoning results are from Gemini API prototyping and a preliminary ablation study on a micro dataset (Section 6.1.5). Full-scale SLM training and evaluation on the complete test set will be supplemented.
3. **VLM backend ablation pending**: A systematic evaluation comparing DePlot, MatCha, Pix2Struct-base, and SVLM backends on the benchmark dataset has not yet been executed. Backend selection is currently based on design rationale (DePlot = chart-specific fine-tuning).

4. **VLM inference latency**: GPU inference for DePlot requires 2–4 seconds per chart. CPU-only deployment is significantly slower. Model distillation or ONNX export is planned for edge deployment.

5. **Domain specificity**: All training data comes from academic papers (arXiv). Generalization to business charts and dashboards is untested.

6. **Vietnamese evaluation pending**: The Vietnamese localization components (VLM prompt configuration, Stage 4 prompt templates, Vietnamese QA training data) are designed but not yet evaluated end-to-end.

6.3 Recommendations

Based on the results and identified limitations, we recommend the following directions for improvement:

1. **Complete SLM fine-tuning and evaluation**: Train and evaluate the Qwen-2.5-1.5B model on the 268,799-sample dataset. Target metrics: JSON valid rate >95%, field accuracy >90%, numeric accuracy >85%, inference latency <2 seconds per chart.

2. **VLM backend ablation study**: Evaluate DePlot, MatCha, Pix2Struct-base, and SVLM backends systematically on the 50-chart benchmark to select the optimal extractor per chart type. Use the built-in `extractor_backend` configuration to switch backends without code changes.

3. **VLM inference optimization**: Implement batch inference and persistent model loading across chart batches to reduce Stage 3 per-chart latency. Explore ONNX export for CPU deployment.

4. **Vietnamese QA integration**: Configure SVLM backend for Vietnamese output, generate 5,000–10,000 Vietnamese QA samples, fine-tune SLM with combined English + Vietnamese data, and evaluate Vietnamese chart understanding accuracy.

5. **Public benchmark evaluation**: Evaluate on ChartQA and PlotQA to enable direct comparison with DePlot, MatCha, and other published methods.

6. **Domain expansion**: Collect and annotate chart images from Vietnamese business reports and government dashboards to evaluate cross-domain and cross-language generalization.

7. **Dataset release**: Prepare the training dataset (English + Vietnamese) and model weights for public release to support community research.

7 Conclusion

7.1 Summary of Contributions

This project designed and implemented **Geo-SLM**, a hybrid neuro-symbolic system for extracting structured numerical data from chart images. The key contributions are:

1. **Hybrid 5-stage pipeline architecture**: A modular pipeline combining deep learning (YOLO, ResNet-18), classical geometry (RANSAC, RDP, skeleton analysis), and AI reasoning (SLM with fallback to cloud APIs). Each stage operates on validated Pydantic schema contracts, enabling independent development, testing, and replacement.

2. **Geometric calibration for chart value extraction**: RANSAC-based axis calibration with pixel-to-value mapping achieves deterministic, traceable value extraction. This fundamentally differs from end-to-end models that estimate values from pixel patterns.

3. **Large-scale academic chart dataset**: 32,364 classified charts from ∼4,000 arXiv papers, with structured geometric features (axis calibration, OCR with spatial roles, element detection). The derived SLM training dataset contains 268,799 ChatML conversation samples across 8 chart types.

4. **Language-extensible architecture for Vietnamese chart Q&A**: The pipeline is designed with a Core-first, Localize-second strategy where 12 out of 17 components are fully language-agnostic. Vietnamese support is achieved through configuration changes (OCR language), additive extensions (keyword dictionaries, prompt templates), and targeted data generation (Vietnamese QA samples) – without modifying any core pipeline logic.

5. **Comprehensive evaluation**: Chart detection at 93.5% mAP@0.5, classification at 94.14% accuracy, and 100% batch extraction success rate across the full 32,364-chart corpus with 92.6% overall pipeline confidence.

7.2 Achievement of Objectives

Table 21: Achievement of project objectives.

Objective	**Target**	**Achieved**	**Status**
Detection mAP@0.5	$> 85\%$	93.5%	Exceeded
Classification accuracy	$> 90\%$	94.14%	Exceeded
Extraction confidence	$> 90\%$	92.6%	Exceeded
Training dataset size	$> 200{,}000$	268,799	Exceeded
Consumer GPU deployment	RTX 3060	Confirmed	Met
Vietnamese Q&A support	End-to-end	Architecture ready	In progress
SLM fine-tuning	Complete	In progress	Pending
Public benchmark eval	ChartQA/PlotQA	Not yet	Pending

Five of seven objectives were exceeded. The Vietnamese Q&A architecture is in place (12/17 components language-agnostic, localization layers designed), pending QA data generation and SLM fine-tuning. The remaining items are in progress.

7.3 Future Work

1. **Vietnamese chart Q&A deployment**: Execute the full Vietnamese localization plan: configure OCR for Vietnamese diacritical marks, generate 5,000–10,000 Vietnamese QA samples via Data Factory, fine-tune SLM with combined English + Vietnamese data, and evaluate Vietnamese chart understanding accuracy end-to-end.

2. **SLM fine-tuning and evaluation**: Complete QLoRA training of Qwen-2.5-1.5B on the 268,799-sample dataset (English) plus Vietnamese supplement. Target: JSON valid rate >95%, field accuracy >90%, Vietnamese output fluency >90%, latency <2s per chart.

3. **Public benchmark comparison**: Evaluate on ChartQA [13] and PlotQA [14] for direct comparison with DePlot [1] and MatCha [2].

4. **Ablation study**: Systematically measure the contribution of the negative image transform, adaptive RDP vectorization, RANSAC calibration, and SLM reasoning through controlled experiments.

5. **OCR optimization**: Implement batch processing and selective re-execution to reduce Stage 3 latency from ∼5.0s to <2.5s.

6. **Vietnamese domain expansion**: Collect chart images from Vietnamese government reports, business dashboards, and academic publications to build a Vietnamese chart benchmark dataset.

7. **Production deployment**: Integrate the Celery task queue and FastAPI server for asynchronous pipeline execution with PostgreSQL state management.

8. **Dataset release**: Prepare the training dataset (English + Vietnamese) and model weights for public release to support community research.

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